

PHY1004W (PSALE010 WPS10

$$1a) \langle 1, -5, -3 \rangle \cdot \langle 4, 2, -7 \rangle = (1)(4) + (-5)(2) + (-3)(-7) \\ = 15u^2$$

$$b) \langle 4, 2, -7 \rangle \times \langle 1, -5, -3 \rangle = \langle \cancel{(2)(-3)} - \cancel{(-7)(-5)}, \langle (2)(-3) - (-7)(-5), (-7)(1) - (4)(-3), (4)(-5) - (2)(1) \rangle \\ = \langle -4, 5, -22 \rangle u^2$$

$$c) \langle 1, -5, -3 \rangle \times \langle 4, 2, -7 \rangle \\ = \langle (-5)(-7) - (-3)(2), (-3)(4) - (1)(-7), (1)(2) - (-5)(4) \rangle \\ = \langle 41, -5, 22 \rangle u^2$$

$$2. \vec{L}_A = \vec{r}_A \times \vec{p} \\ = \langle 6, 6, 0 \rangle \times \langle -11, 13, 0 \rangle \\ = \langle (6)(0) - (0)(13), (0)(-11) - (6)(0), (6)(13) - (6)(-11) \rangle \\ = \langle 0, 0, 144 \rangle \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$3. (1) L_{Az} = r_x p_y - r_y p_x \\ = (-w)(0) - (h)(25) \\ = -28 \cdot 25 = -700 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$(2) L_{Az} = r_x p_y - r_y p_x \\ = (0)(0) - (h)(25) = -700 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$(3) L_{Az} = r_x p_y - r_y p_x = -700 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$(4) L_{Az} = r_x p_y - r_y p_x \\ = (-w)(25) - (0)(0) = -18 \cdot 25 = -450 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$(5) L_{Az} = r_x p_y - r_y p_x \text{ and } \vec{r} = \vec{0} \text{ so } L_{Az} = 0 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$(6) L_{Az} = r_x p_y - r_y p_x \\ = (w)(0) - (0)(25) = 0 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$(7) \text{ Since } \vec{L}_A = \vec{r}_A \times \vec{p} \Rightarrow \hat{L}_A = \langle 0, 0, -1 \rangle \text{ by the right hand thumb rule and thus } L_{Az} = -|\vec{L}_A| \\ = -r p \sin \theta = -27 \cdot 25 \sin 90 = -675 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$4. (a) \vec{p} = m\vec{v}$$

$$= 40 \langle 7.5, 0, 0 \rangle$$

$$= \langle 300, 0, 0 \rangle \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}$$

(b)  $\vec{r} = \langle 0, 24, 0 \rangle \text{ m}$  since this is the displacement from the child to the position A to the child

$$(c) \vec{r}_A = \langle 0, 24, 0 \rangle \text{ m}$$

$$(d) |\vec{L}_A| = |\vec{r}_A| |\vec{p}| \sin \theta = 24 \times 300 = 7200 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

(e) The  $xy$ -plane

(f) The ferris wheel moves anti-clockwise ( $v_x > 0$  at F) and thus  $\hat{L}_A = \langle 0, 0, 1 \rangle$

$$\therefore L_{Az} = 1(7200) = 7200 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$(g) L_{Az} = r_x p_y - r_y p_x$$

$$= (0)(0) - (-24)(300) = 7200 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$(h) |\vec{r}_A| = 24 \text{ and } |\vec{p}| = m|\vec{v}|$$

$$= 40 \sqrt{(5.303)^2 + (5.303)^2}$$

$$= 300 \text{ kg} \cdot \text{m} \cdot \text{s}^{-1}$$

$$\therefore |\vec{L}_A| = |\vec{r}_A| |\vec{p}| \sin \theta = 24 \cdot 300$$

$$= 7200 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$$

$$5. (a) I = \frac{1}{2} MR^2$$

$$= \frac{1}{2} (13)(0.2^2) = 0.26$$

$$\text{Now } \omega = \frac{2\pi}{T} = \frac{2\pi}{0.6} = 10.47 \text{ rad} \cdot \text{s}^{-1}$$

$$\text{Thus } |\vec{L}_{\text{rot}}| = I\omega = 2.72 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-1}$$

Now since the rotation is clockwise this

$$\text{means } \hat{L}_{\text{rot}} = \langle 0, 0, -1 \rangle \Rightarrow \vec{L}_{\text{rot}} = \langle 0, 0, -2.72 \rangle$$

$$K_{\text{rot}} = \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} (0.26)(10.47^2) = 14.25 \text{ J}$$

$$(b) I = \frac{2}{5} MR^2$$

$$= \frac{2}{5} (22)(0.7^2) = 4.312$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.5} = 12.57 \text{ rad} \cdot \text{s}^{-1}$$

Using the right hand thumb rule (with anti-clockwise rotation), ~~you~~  $\hat{L}_{\text{rot}} = \langle 1, 0, 0 \rangle$

Now  $|\vec{L}_{\text{rot}}| = I\omega = (4.312)(12.57) = 54.19$

$\Rightarrow \vec{L}_{\text{rot}} = \langle 54.19, 0, 0 \rangle \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-1}$

$K_{\text{rot}} = \frac{1}{2} I \omega^2 = \frac{1}{2} (4.312)(12.57)^2 = 340.66 \text{ J}$

6. (a)  $I = \cancel{\frac{1}{2} M R^2} M_1 r_{11}^2 + M_2 r_{12}^2 + \dots + M_n r_{1n}^2$   
 $\cancel{= \frac{1}{2} (10.82)(0.32)^2} = 2(0.82(0.16)^2 + 0.29(0.08)^2)$   
 $\cancel{= 0.0219} = 0.0457$

(b)  $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.9} = 6.98 \text{ rad} \cdot \text{s}^{-1}$

(c)  $L_{\text{rot}} = I\omega$   
 $= 0.0457(6.98)$   
 $= 0.319 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-1}$

7. (a)  $L_{\text{trans, sun}} = r p \sin \theta$   
 $= (1.5 \times 10^{11}) (M_E v_E) \sin 90^\circ \text{ (orbit circular)}$

Now  $T = 365 \text{ days}$  so  $v = \frac{2\pi r}{T}$   
 $= \frac{2\pi(1.5 \times 10^{11})}{365 \times 24 \times 60 \times 60} = 29885.77$

$\Rightarrow L_{\text{trans, sun}} = (1.5 \times 10^{11})(6 \times 10^{24} \times 29885.77) = 2.7 \times 10^{40}$

(b)  $I = \frac{2}{5} M R^2$   
 $= \frac{2}{5} (6 \times 10^{24})(6.4 \times 10^6)^2 = 9.83 \times 10^{37}$

$\omega = \frac{2\pi}{T} = \frac{2\pi}{24 \times 60 \times 60} = 7.27 \times 10^{-5} \text{ rad} \cdot \text{s}^{-1}$

$\therefore L_{\text{rot}} = I\omega = (9.83 \times 10^{37})(7.27 \times 10^{-5})$   
 $= 7.15 \times 10^{33} \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-1}$